A Conceptual Guide to Global SU(2) Heat-Kernel Action Enclosures

Geometry, diffusion, lattice gauge theory, and why derivative bounds matter

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Abstract

This document is a pedagogical companion to the paper *Global Arbitrary-Order Enclosures for the* SU(2) *Heat-Kernel Action*. It is written for a first-year PhD student who knows basic quantum field theory or differential geometry but has not yet worked with compact-group heat kernels. The central object is the negative logarithm of the heat kernel on SU(2), used as a one-plaquette action in lattice gauge theory.

The guide develops the complete conceptual chain. A continuum gauge field produces link holonomies; their ordered product around a plaquette is an element of SU(2); conjugation reduces that element to a class angle; diffusion on the group supplies a normalized probability density; and its negative logarithm supplies an action. Two exact representations of the heat kernel then reveal complementary structures: the character expansion makes normalization and convolution transparent, while the Poisson-image formula exhibits the short-time Gaussian. Their apparent endpoint singularities lead naturally to a three-chart analysis. Near the antipode, a continuous S^2 family of minimizing geodesics changes the short-time prefactor and the Hessian behavior. We explain why derivative bounds of arbitrary prescribed order, positivity estimates, Haar-probability tails, and exact rational validation are useful inputs for constructive and renormalization-group analysis. We also separate these local results from the further arguments needed for a continuum Yang–Mills theory or a mass gap.

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1 The broader picture

Yang–Mills theory describes gauge fields: geometrically, connections on principal bundles; physically, the carriers of nonabelian interactions. In four Euclidean dimensions the formal continuum action is

$$S_{\mathrm{YM}}[A] = \frac{1}{4g_0^2} \int \sum_{a,\mu,\nu} F^a_{\mu\nu}(x) F^a_{\mu\nu}(x) \, \mathrm{d}^4x, \tag{1}$$

where A is a gauge potential, F is its curvature, and g_0 is a coupling constant. This compact formula conceals the main mathematical difficulty: the desired quantum theory is an infinite-dimensional probability measure modulo gauge symmetry, and the physically relevant infrared behavior is strongly nonperturbative.

Lattice gauge theory replaces continuous space by a grid and the connection by group-valued parallel transports on links. This gives a finite-dimensional regulated model. The regulator does not solve the theory, but it makes the basic objects honest integrals rather than formal functional integrals. The smallest gauge-invariant observable on the lattice is the plaquette holonomy, the ordered product of four link variables around an elementary square. A lattice action assigns an energetic cost to each plaquette holonomy.

The heat-kernel action is one particularly natural choice. Instead of choosing an arbitrary class function on the gauge group, one uses the negative logarithm of the group's diffusion kernel. The semigroup property of diffusion then mirrors the composition of scales. This is why heat-kernel actions have long been important in lattice gauge theory [MO81, Man80, BDJ85].

The paper addresses one link in this much longer chain: once the one-plaquette heat-kernel action has been chosen, can we bound its derivatives explicitly, at any requested finite order, across the entire compact group? This is a local and finite-dimensional question, but it is the kind of input

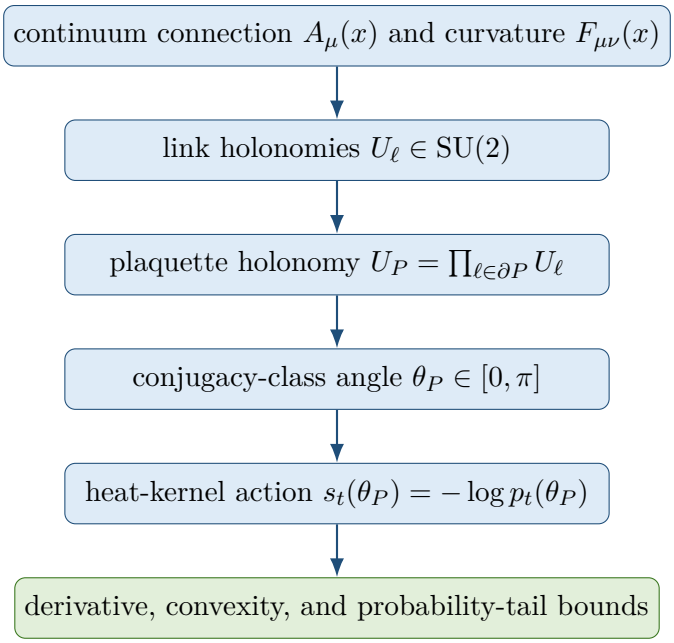


Figure 1: The chain from continuum gauge geometry to the quantitative object studied in the paper. Each arrow introduces a convention that must be fixed before numerical constants can be compared.

required before one can control nonlinear changes of variables, Taylor remainders, or renormalization steps.

The central question

For the exactly normalized heat kernel on SU(2), how can one obtain explicit finite-time bounds for derivatives of $s_t = -\log p_t$ from the identity all the way to the cut locus, without hiding singular denominators or relying only on asymptotic notation?

2 Minimum gauge-theory background

2.1 Connections, holonomy, and gauge invariance

A gauge potential $A_\mu(x)$ is locally a Lie-algebra-valued one-form. Given a path γ, parallel transport along the path produces a group element

$$U_\gamma = \mathcal{P}\exp\left(i\int_\gamma A\right),$$

where $\mathcal{P}$ orders noncommuting factors along the path. Under a gauge transformation, an open-path holonomy transforms at its endpoints. The holonomy around a closed loop transforms by conjugation, so its trace and conjugacy class are gauge invariant.

On a hypercubic lattice, the fundamental variables are link matrices $U_{x,\mu} \in \mathrm{SU}(2)$. For an elementary plaquette in the $\mu\nu$ plane,

$$U_P = U_{x,\mu}U_{x+\hat{\mu},\nu}U^{-1}_{x+\hat{\nu},\mu}U^{-1}_{x,\nu}. \tag{2}$$

The total action is normally a sum of class functions of U_P. The one-plaquette analysis asks how one such class function behaves before the plaquettes are coupled through their shared links.

2.2 Why start with SU(2)?

SU(2) is the smallest nonabelian compact Lie group. It already contains noncommuting gauge degrees of freedom, yet its geometry is unusually explicit: as a Riemannian manifold with the standard bi-invariant metric, it is the unit three-sphere S^3.

Using the Pauli matrices σ^a and generators $T^a = \sigma^a/2$, every group element can be written as

$$U = \cos\theta\,\mathbf{1} + i\sin\theta\,\hat{n}\cdot\sigma = \exp\bigl(2i\theta\,\hat{n}\cdot T\bigr), \qquad 0 \le \theta \le \pi, \tag{3}$$

for a unit vector $\hat{n} \in S^2$. Conjugation rotates $\hat{n}$ but does not change θ. Consequently every central, or conjugation-invariant, function depends only on the class angle θ.

The basic identities are

$$\frac{1}{2}\operatorname{Tr} U = \cos\theta, \qquad d(\mathbf{1}, U) = \theta, \qquad |X| = 2\theta \quad \text{if } U = \exp(iX^aT^a). \tag{4}$$

The final equality is an important convention warning: the norm of the physics Lie-algebra coordinate X is twice the geodesic class angle used in the heat-kernel formulas.

Normalized Haar probability reduces on central functions to

$$\int_{\mathrm{SU}(2)} f(U)\ \mathrm{d}H(U) = \frac{2}{\pi}\int_0^\pi f(\theta)\sin^2\theta\ \mathrm{d}\theta. \tag{5}$$

The factor $\sin^2\theta$ is the radial volume element of S^3. It vanishes at both endpoints because the two-sphere of angular directions collapses at the identity and at the antipode.

Definition 2.1 (Class angle)**.** The class angle $\theta \in [0, \pi]$ is the unique number satisfying $\frac{1}{2}\operatorname{Tr} U = \cos\theta$. It is a coordinate on the space of conjugacy classes, not a global coordinate system on the three-dimensional group.

3 Heat kernels: diffusion made geometric

3.1 From Euclidean diffusion to a compact group

In flat d-dimensional Euclidean space, the heat equation $\partial_t u = \frac{1}{2}\Delta u$ has fundamental solution

$$k_t(x, y) = (2\pi t)^{-d/2}\exp\left[-\frac{|x-y|^2}{2t}\right]. \tag{6}$$

It is simultaneously a diffusion transition density and a normalized Gaussian. On a Riemannian manifold, the heat kernel $p_t(x, y)$ plays the same role, but curvature, topology, and multiple geodesics modify the flat formula.

For a compact Lie group with a bi-invariant metric, the heat kernel starting at the identity is central. We write it as $p_t(U)$ or $p_t(\theta)$ and fix the convention

$$\partial_t p_t = \frac{1}{2}\Delta p_t, \qquad \lim_{t\downarrow 0} p_t = \delta_{\mathbf{1}}, \qquad \int_{\mathrm{SU}(2)} p_t\ \mathrm{d}H = 1. \tag{7}$$

Here t is heat time. In the lattice application it is related to a bare coupling, but it is not Lorentzian time and should not be interpreted as the physical duration of a process.

The semigroup law is

$$p_{t_1+t_2}(U) = \int_{\mathrm{SU}(2)} p_{t_1}(UV^{-1})p_{t_2}(V)\ \mathrm{d}H(V). \tag{8}$$

It says that two successive diffusion steps compose by group convolution. This exact scale-composition property is the structural attraction of the heat-kernel action.

3.2 Why take the negative logarithm?

A probability weight in Euclidean field theory has the form e^{-S}. If a single plaquette is weighted by $p_t(U_P)$, its action is therefore

$$s_t(U_P) = -\log p_t(U_P). \tag{9}$$

Near the identity and at short heat time, the diffusion kernel is approximately Gaussian in geodesic distance, so

$$s_t(\theta) \approx \frac{\theta^2}{2t} + \text{curvature corrections} + \text{image corrections} + \text{constant}. \tag{10}$$

Thus s_t resembles the Manton geodesic action, with systematic compact-group corrections. Additive constants affect the partition function but not action derivatives.

Guiding principle 3.1 (Probability and action viewpoints)**.** *The heat kernel p_t is the natural object for normalization, convolution, and tail probabilities. The action $s_t = -\log p_t$ is the natural object for forces, Hessians, Taylor expansions, and convexity. Passing from one to the other is safe only after a strictly positive lower bound for p_t has been established.*

4 Two exact formulas, two kinds of insight

The same SU(2) heat kernel admits two exact representations. The entire paper is driven by the fact that neither one is uniformly convenient in all regimes.

4.1 The character expansion

Irreducible representations of SU(2) are labelled by $n \in \mathbb{N}_0$ or spin $j = n/2$. The representation has dimension $n+1$, character

$$\chi_n(\theta) = \frac{\sin((n+1)\theta)}{\sin\theta},$$

and Laplacian eigenvalue $-n(n+2)$. Spectral decomposition gives

$$p_t(\theta) = \sum_{n=0}^{\infty} (n+1)\mathrm{e}^{-n(n+2)t/2} \frac{\sin((n+1)\theta)}{\sin\theta}. \tag{11}$$

This formula makes several facts immediate:

- the kernel is built from group representations;
- integration against Haar probability removes every nontrivial character, so normalization is exact;
- large representations are exponentially damped at fixed positive t;
- the convolution law follows from the multiplication of spectral weights.

In spin notation the weight is $\mathrm{e}^{-2tj(j+1)}$. A source that writes the weight as $\mathrm{e}^{-\tau C_2(j)}$ is therefore using $\tau = 2t$ under the present generator convention.

4.2 The Poisson-image formula

Poisson summation reorganizes the same kernel into a sum over radial images:

$$p_t(\theta) = C_t \frac{1}{\sin\theta} \sum_{k\in\mathbb{Z}} (\theta + 2\pi k) \mathrm{e}^{-(\theta+2\pi k)^2/(2t)}, \qquad C_t = \frac{\mathrm{e}^{t/2}\sqrt{2\pi}}{2t^{3/2}}. \tag{12}$$

The $k = 0$ term is the local Gaussian. The $k \neq 0$ terms encode the compact topology through paths that wrap around the radial covering coordinate. At small t, their Gaussian suppression makes the short-distance structure visible with very few terms.

Representation	What it makes transparent	Where care is needed
Character sum	Haar normalization, representation content, semigroup law, moderate and large heat time	many spectral modes can be needed at very small heat time
Poisson images	short-time Gaussian, exponentially small images, endpoint pairing, explicit tail estimates	the factor $1/\sin\theta$ is singular term by term at 0 and π

Table 1: The two exact representations answer different questions.

The formulas are exactly equal, rather than two unrelated approximations. This equality lets one choose the representation that best matches the geometric regime.

5 Why the endpoints are subtle

5.1 A removable singularity is still dangerous in a proof

The factor $1/\sin\theta$ in (12) diverges at both endpoints. The full heat kernel is smooth, so the divergence must cancel in the complete numerator. But a smooth final answer does not justify differentiating singular pieces separately. Truncation or interval bounds can destroy the cancellation unless the relevant images are paired first.

Near the identity, pair the images at $+2\pi k$ and $-2\pi k$. Factoring out the zero image gives

$$p_t(\theta) = C_t \mathrm{e}^{-\theta^2/(2t)} \frac{\theta}{\sin\theta} (1 + q_t(\theta)), \tag{13}$$

where

$$q_t(\theta) = 2 \sum_{k\geq 1} \mathrm{e}^{-(2\pi k)^2/(2t)} \left[\cosh\frac{2\pi k\theta}{t} - \frac{2\pi k}{\theta} \sinh\frac{2\pi k\theta}{t} \right]. \tag{14}$$

The quotient involving θ has an even Taylor expansion. An integral identity removes it before differentiation:

$$\frac{a}{\theta}\sinh\frac{a\theta}{t} = \frac{a^2}{t}\int_0^1 \cosh\frac{sa\theta}{t}\,\mathrm{d}s. \tag{15}$$

This is a small technical step with a large conceptual consequence: all derivatives can now be bounded using one nonsingular expression.

Taking the logarithm of (13) yields the exact decomposition

$$s_t(\theta) = \frac{\theta^2}{2t} + \log\frac{\sin\theta}{\theta} - \log(1 + q_t(\theta)) - \log C_t. \tag{16}$$

The four terms have distinct meanings:

(i) $\theta^2/(2t)$ is the flat Gaussian action;

(ii) $\log(\sin\theta/\theta)$ is the local curvature or volume correction;

(iii) $-\log(1+q_t)$ is the compact-topology image correction;

(iv) $-\log C_t$ fixes normalization and disappears under spatial derivatives.

5.2 The antipode and the cut locus

The antipode $\theta=\pi$ is geometrically different from the identity. On S^3, every unit tangent direction at the identity determines a length-π geodesic ending at the antipode. The minimizing geodesics therefore form an S^2 family.

This statement must be separated from the one-dimensional image formula. The two leading terms in that formula have radial arguments approaching $+\pi$ and $-\pi$. They are two affine-Weyl image terms, not two individual geodesics. Their paired limit nevertheless encodes the continuous geometric degeneracy.

With $\phi=\pi-\theta$ and $b_j=(2j+1)\pi$, the paired numerator is

$$2\mathrm{e}^{-(b_j^2+\phi^2)/(2t)}\left[b_j\sinh\frac{b_j\phi}{t}-\phi\cosh\frac{b_j\phi}{t}\right]. \tag{17}$$

Pairing occurs before division by $\sin\phi$. The exact endpoint sum is

$$p_t(\pi)=2C_t\sum_{j\geq 0}\mathrm{e}^{-(2j+1)^2\pi^2/(2t)}\left(\frac{(2j+1)^2\pi^2}{t}-1\right). \tag{18}$$

Its leading term gives

$$p_t(\pi)\sim \mathrm{e}^{t/2}\sqrt{2\pi}\,\pi^2 t^{-5/2}\mathrm{e}^{-\pi^2/(2t)}\qquad (t\downarrow 0). \tag{19}$$

Away from a cut locus, a three-dimensional heat kernel has the generic power $t^{-3/2}$. A clean two-dimensional family of minimizing geodesics adds an extra $t^{-2/2}=t^{-1}$ enhancement, producing $t^{-5/2}$. This is the Morse–Bott mechanism. General heat-kernel results explain this phenomenon in much broader settings [CLW23, NS25]; the SU(2) image sum makes it completely explicit.

Geometric lesson

The exponent $\exp[-d^2/(2t)]$ records the shortest geodesic distance. The power of t records how fluctuations accumulate around the minimizing set. At the antipode the distance is still π, but the set of minimizers is two-dimensional, so the prefactor changes.

6 The three-chart strategy

No single estimate is both simple and well conditioned across the whole class interval. The paper therefore uses a small atlas adapted to the geometry:

$$[0,\rho]\quad\text{identity chart},\qquad[\rho,\pi-\delta]\quad\text{transition chart},\qquad[\pi-\delta,\pi]\quad\text{cut chart}.$$

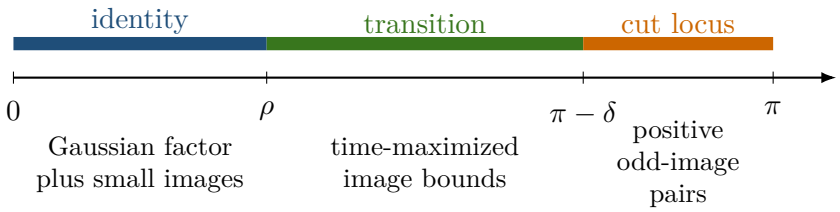


Figure 2: A formula-adapted atlas on the conjugacy-class interval.

6.1 Identity chart: perturbation of a Gaussian

On $[0, \rho]$, the image correction q_t in (13) is exponentially small at short heat time. For each derivative order m, the paper constructs a computable number Q_m such that

$$\sup_{0<t\le t_0} \sup_{0\le\theta\le\rho} \left| q_t^{(m)}(\theta) \right| \le Q_m. \tag{20}$$

The crucial zeroth-order condition is $Q_0 < 1$. It implies

$$1 + q_t(\theta) \ge 1 - Q_0 > 0, \tag{21}$$

so the logarithm and its derivatives have a controlled denominator.

The curvature term has an independently convergent expansion from Euler's product for the sine:

$$\log \frac{\sin\theta}{\theta} = -\sum_{n\ge1} \frac{\zeta(2n)}{n\pi^{2n}} \theta^{2n}, \qquad |\theta| < \pi. \tag{22}$$

Thus the Gaussian, geometric, and image contributions are bounded separately and then recombined.

6.2 Transition chart: remove an artificial order restriction

The simplest identity-chart bound replaces the heat-time dependence by its value at t_0. This is valid only when each exponential–polynomial factor is monotone on $(0, t_0]$, and the condition becomes stricter as the derivative order grows.

The transition chart instead maximizes the relevant function exactly:

$$\sup_{0<t\le t_0} \mathrm{e}^{-A/t} t^{-r} = \begin{cases} \mathrm{e}^{-A/t_0}, & r = 0, \\ \mathrm{e}^{-A/\tau_r} \tau_r^{-r}, & r \ge 1, \end{cases} \qquad \tau_r = \min\{t_0, A/r\}. \tag{23}$$

This elementary optimization matters because spatial differentiation produces powers of t^{-1}. Handling the maximum exactly gives uniform finite-time bounds without pretending the endpoint t_0 is always the worst case.

6.3 Cut chart: bound the kernel before its logarithm

Near π, the zero-image factorization is poorly conditioned because the zero image is no longer the unique dominant geometric contribution. The paper instead works with the positive odd-image pairs (17). It constructs

- derivative bounds $D_m^{\mathrm{cut}}(t, \delta)$ for p_t itself;

- a strictly positive principal-pair lower bound $L_{\mathrm{cut}}(t,\delta)$ for p_t;
- logarithmic derivative bounds obtained from the ratio of these quantities.

The cut-chart constants necessarily depend on t. This is not a weakness of the estimate. In fact, the antipodal action Hessian behaves as

$$s_t''(\pi) = -\frac{\pi^2}{3t^2} + O(t^{-1}), \tag{24}$$

so no uniform $t \downarrow 0$ Hessian bound can hold there.

7 How derivative bounds pass through a logarithm

Suppose a positive function has the form $f = 1 + q$ and we know bounds for $q, q', \ldots, q^{(M)}$. The first few derivatives illustrate the pattern:

$$\begin{aligned}(\log f)' &= \frac{f'}{f},\\ (\log f)'' &= \frac{f''}{f} - \frac{(f')^2}{f^2},\\ (\log f)''' &= \frac{f'''}{f} - 3\frac{f'f''}{f^2} + 2\frac{(f')^3}{f^3}.\end{aligned}$$

Every order contains a structured sum of products. Writing each order by hand is error prone. Faà di Bruno's formula organizes the result through exponential partial Bell polynomials:

$$\partial_\theta^m \log f = \sum_{\ell=1}^{m} (-1)^{\ell-1}(\ell-1)! \frac{\mathsf{B}_{m,\ell}(f', \ldots, f^{(m-\ell+1)})}{f^\ell}. \tag{25}$$

If $f \geq L > 0$ and $|f^{(j)}| \leq D_j$, this immediately gives an explicit upper bound at every requested finite order.

Guiding principle 7.1 (Why the zeroth-order bound comes first)**.** *Derivative estimates for p_t alone do not control derivatives of $-\log p_t$. One must first prove $p_t \geq L > 0$ on the domain. The lower bound is the bridge between a heat-kernel estimate and an action estimate.*

8 What "arbitrary order" means

The result is arbitrary-order in an effective, finite sense. Given any chosen $M < \infty$, the construction outputs bounds through order M together with explicit convergent tails. It does not assert that one fixed small-time domain gives a uniform constant for all M simultaneously.

This distinction is standard and important. High derivatives introduce large powers of $1/t$ and increasing combinatorial factors. A statement useful for computation or renormalization is therefore naturally organized as

$$\text{choose } M \quad \longrightarrow \quad \text{choose a compatible domain} \quad \longrightarrow \quad \text{compute constants through order } M.$$

Why might one need more than the Hessian?

1. The first derivative gives the local force in a variational or sampling algorithm.

2. The Hessian gives convexity, stiffness, and Gaussian comparison.

3. Third and fourth derivatives control nonlinear Taylor remainders and the leading corrections to a quadratic approximation.

4. Higher derivatives enter repeated changes of variables, cumulant expansions, and multiscale effective actions.

The value of the theorem is therefore not that every calculation will use order eight or order twenty. It is that the proof method does not have to be redesigned when the required order changes.

9 Hessian geometry and a concrete plaquette estimate

For a radial function $f(\theta)$ on the unit three-sphere, the Hessian has one radial and two tangential eigenvalues:

$$\lambda_{\mathrm{rad}} = f''(\theta), \qquad \lambda_{\tan} = f'(\theta)\cot\theta \quad \text{with multiplicity two.} \tag{26}$$

The factor $\cot\theta$ is geometric. Tangential directions correspond to changing the axis $\widehat{n}$ in (3) while keeping the radial class angle fixed to first order.

If

$$\left|\partial_\theta^2(s_t - \theta^2/(2t))\right| \le R_2 \quad \text{on } [0,\rho],$$

then

$$\lambda_{\mathrm{rad}}(s_t) \ge t^{-1} - R_2, \qquad \lambda_{\tan}(s_t) \ge \rho\cot\rho\, t^{-1} - R_2. \tag{27}$$

These inequalities express a simple physical picture: near the identity the strong Gaussian curvature $1/t$ dominates the compact-geometry and image corrections.

For the validated instance

$$t_0 = \frac{1}{5}, qquad\rho = 1, qquad M = 8,$$

the paper obtains $R_2 < 0.455$. Hence, for every $0 < t \le 1/5$ and $0 \le \theta \le 1$,

$$\lambda_{\mathrm{rad}}(s_t) > 4.545, \qquad \lambda_{\tan}(s_t) > 2.755. \tag{28}$$

At $t = 1/5$, the scaled corrections also satisfy

$$tR_2 < 0.091, \qquad tR_3 < 0.058, \qquad tR_4 < 0.101. \tag{29}$$

Example 9.1 (Commuting radial plaquette path)**.** *Consider*

$$U_P(x) = \exp\bigl(2i(\theta + x)T^3\bigr).$$

Its class angle is $\theta_P(x) = \theta + x$, so no noncommutative chain-rule terms appear. Equation (28) *gives a second derivative at least* 4.545 *throughout the stated chart, while* (29) *controls the third- and fourth-order departure from the Gaussian action. This is a concrete one-plaquette coercivity statement.*

For a general link variation, θ_P is a nonlinear function of several noncommuting link matrices. Then

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2} s_t(\theta_P(x)) = s_t''(\theta_P)(\theta_P')^2 + s_t'(\theta_P)\theta_P''. \tag{30}$$

The paper controls the derivatives of s_t; a full lattice estimate must also control the derivatives of the plaquette map and sum the coupled contributions in a specified norm.

10 Connection with the weak-field continuum action

For a smooth gauge field and lattice spacing a, the plaquette expansion has the form

$$X_P^a = a^2 g_0 F_{\mu\nu}^a(x) + O(a^3), \qquad U_P = \exp(iX_P^a T^a). \tag{31}$$

Since $\theta_P = |X_P|/2$, the Gaussian part of the heat-kernel action gives

$$\frac{\theta_P^2}{2t} = \frac{a^4 g_0^2}{8t} F_{\mu\nu}^a F_{\mu\nu}^a + O\left(\frac{a^5}{t}\right). \tag{32}$$

Summing over plaquettes converts the factor a^4 into a Riemann sum for a four-dimensional integral. This explains how the lattice action can match (1) after the convention-dependent relation between t and the bare coupling is fixed.

The next term in (16) begins as

$$\log \frac{\sin\theta}{\theta} = -\frac{\theta^2}{6} + O(\theta^4).$$

It is a compact-geometry correction without the leading $1/t$ enhancement. The nonzero images are exponentially small on a fixed identity chart as $t \downarrow 0$. Thus the local weak-field picture has a transparent hierarchy:

Gaussian continuum term + group-curvature correction + global image correction.

This matching is a necessary consistency check, but it is not by itself a continuum-limit proof. Such a proof must control the collective probability measure as the number of lattice degrees of freedom diverges and the bare parameters are tuned.

11 Haar tails and the small-field/large-field split

Let $\Pr_t$ be the probability measure on SU(2) with density p_t relative to normalized Haar measure. A tail probability

$$\Pr_t\{\theta \geq \rho\}$$

measures how much of the one-plaquette heat-kernel distribution lies outside the identity chart.

The exact normalization from the character formula avoids estimating a denominator. Combining the image formula with the Haar factor $\sin^2\theta$ cancels one power of the apparent Weyl denominator. The paper obtains an explicit Gaussian-type upper bound whose dominant rate is

$$\log \Pr_t\{\theta \geq \rho\} \lesssim -\frac{\rho^2}{2t} \qquad (t \downarrow 0,\ \rho < \pi). \tag{33}$$

This is relevant to a common constructive strategy:

1. treat the small-field region using convexity and Taylor estimates;
2. show that the large-field region has sufficiently small probability;
3. prove that both controls survive the interaction between neighboring degrees of freedom and successive scale transformations.

The theorem supplies the one-variable tail in step 2. In a four-dimensional lattice measure, plaquettes share links and are not independent. Therefore the one-plaquette tail cannot simply be multiplied across plaquettes. A dependent-measure estimate remains necessary.

12 Why exact rational validation is useful

The analytic theorem produces convergent series and explicit formulas, but a concrete parameter choice still involves many numerical comparisons: Is the factor floor positive? Is an omitted-tail ratio below one? Is the Hessian margin strictly positive after all upper bounds are combined?

The accompanying computation stores theorem-critical quantities as rational numbers and decides strict inequalities using integers. Floating-point evaluation is reserved for a separate comparison of the character and image formulas. For the instance $t_0 = 1/5$, $\rho = 1$, $\delta = 1/5$, and $M = 8$, the record includes

- identity-chart image, logarithmic, and geometric bounds;
- transition-chart bounds through order eight;
- the cut-pair positivity margin;
- normalized tail bounds;
- radial and tangential plaquette Hessian margins.

This computational layer serves three purposes. It prevents roundoff from deciding a strict inequality, makes every numerical constant reproducible, and catches normalization mistakes by comparing two exact formulas at sample points including both endpoints. It does not replace the analytic proof: the code evaluates and checks a theorem whose inequalities are established in the manuscript.

13 Where this result sits in the Yang–Mills programme

It is useful to distinguish levels of mathematical control. The paper closes the first highlighted level in the diagram below; the later levels require new arguments.

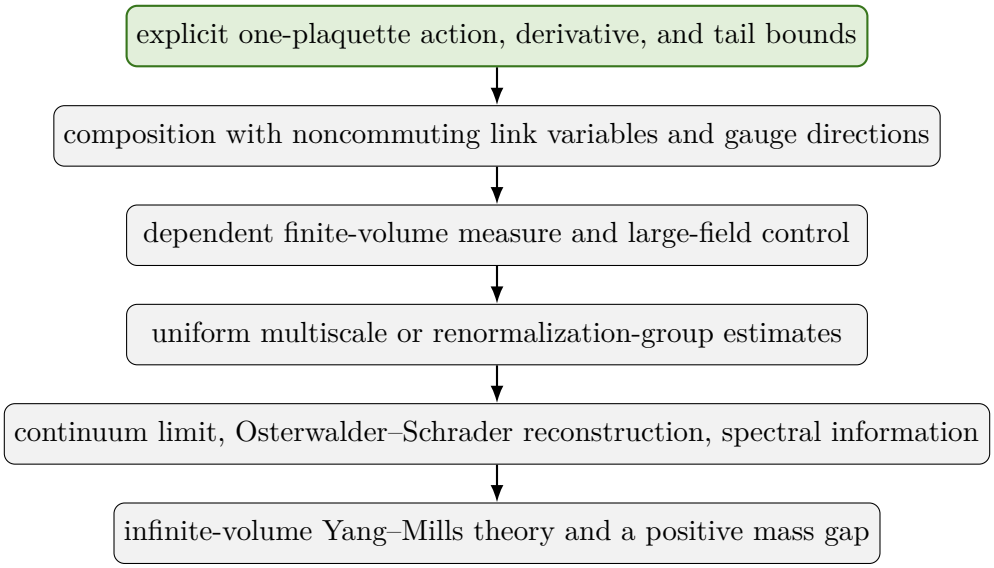


Figure 3: A hierarchy of proof obligations. The local result is useful because later stages need quantitative microscopic input; it does not collapse the later stages into consequences.

The distinction prevents two opposite misunderstandings.

- The result is not a mass-gap proof. It does not construct an infinite-volume interacting measure, prove a continuum limit, or establish a spectral gap for the reconstructed Hamiltonian.
- Beyond the exact heat-kernel formula, the result supplies explicit finite-time, arbitrary-finite-order control, including the cut locus, in a form that can be inserted into later estimates.

The natural next mathematical questions are whether the one-plaquette bounds can be transferred through the noncommutative plaquette map, whether a small-field basin remains stable under block transformations, and whether large-field probabilities can be controlled uniformly in volume.

14 How to read the technical paper

The main paper is easier to navigate once each section is associated with a conceptual question.

Paper component	Question answered
Geometry and normalization	Which metric, generator, Haar measure, class angle, trace, and physics-generator conventions define the constants?
Spectral and image formulas	Which two exact representations expose normalization and short-time geometry?
Image hierarchy	How are paired nonzero images bounded through an arbitrary chosen derivative order near the identity?
Action derivatives	How do those bounds pass safely through the logarithm, and what do they imply for the Hessian?
Global atlas	How are the transition region and antipodal cut locus treated without hiding endpoint singularities?
Haar tails	How much normalized heat-kernel mass lies outside the local small-field chart?
Rational validation	Do the explicit parameter choices satisfy every strict inequality using exact comparisons?
Lattice application	How do the group conventions match a plaquette holonomy, and what concrete convexity margins result?
Appendices	Where are Poisson summation, image tails, Bell polynomials, cut-pair bounds, and endpoint limits proved in detail?

When checking the paper, it is useful to follow one dependency chain rather than reading every formula in isolation:

$$\begin{aligned}\text{paired image formula} &\Longrightarrow \text{derivative bounds} \Longrightarrow \text{positive lower bound} \\ &\Longrightarrow \text{logarithmic derivatives} \Longrightarrow \text{action Hessian and remainders.}\end{aligned} \tag{34}$$

The cut chart follows the same logic with p_t itself in place of the factor $1 + q_t$.

15 Conceptual glossary

Term	Meaning in this work
Class function	A function invariant under $U \mapsto VUV^{-1}$; on SU(2) it depends only on θ.
Class angle	The number $\theta \in [0, \pi]$ determined by $\frac{1}{2} \operatorname{Tr} U = \cos \theta$.
Haar probability	The unique left- and right-invariant probability measure on the compact group.
Heat kernel	The normalized transition density for Brownian diffusion generated here by $\frac{1}{2}\Delta$.
Heat-kernel action	The class function $s_t = -\log p_t$ used as a plaquette action.
Character	The trace of a group representation; characters diagonalize central convolution.
Poisson image	A Gaussian term in the resummed radial covering-space formula; it is not an individual geodesic.
Weyl denominator	The factor $\sin \theta$ appearing in the radial character and image formulas.
Cut locus	Points where minimizing geodesics cease to be unique or the distance function loses smoothness; for the identity on S^3, the cut locus is the antipode.
Morse–Bott family	A smooth positive-dimensional family of stationary or minimizing configurations; here the S^2 family of antipodal geodesics.
Enclosure	An explicit lower or upper bound, often represented by rational endpoints, that is guaranteed to contain the target quantity.
Positivity floor	A proved number $L > 0$ such that the kernel or correction factor is at least L throughout a chart.
Arbitrary finite order	A method that produces bounds for every chosen $M < \infty$, with constants and domain conditions allowed to depend on M.
Bell polynomial	A combinatorial polynomial organizing derivatives of a composition such as $\log f$.
Plaquette	The smallest elementary loop on a lattice and the basic local carrier of curvature.
Small-field chart	A neighborhood of the identity where the Gaussian action dominates its compact-group corrections.

16 Takeaway

The heat-kernel action sits at an instructive intersection of geometry, probability, representation theory, and lattice gauge theory. Its spectral formula says that diffusion is organized by group representations. Its image formula says that short-time diffusion is locally Gaussian but globally aware of compact topology. Its logarithm turns probability into an action. Its endpoint behavior remembers the multiplicity of minimizing geodesics.

The technical achievement of the companion paper is to make these statements quantitative on the full class interval. Pairing removes false endpoint singularities; chart-specific estimates respect the different identity and cut-locus geometries; positivity bounds make logarithmic differentiation legitimate; Bell polynomials extend the calculation to any prescribed finite order; and rational validation turns the analytic estimates into reproducible numerical instances.

For a new researcher, the most important perspective is one of scale and scope. The result provides a controlled microscopic action and a precise small-field input. The next frontier is to

show that this local control survives gauge coupling, large fields, volume growth, and repeated coarse graining. That is where the one-plaquette geometry becomes part of the wider constructive Yang–Mills problem.

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